

Sijil — Product Requirements Document

Version: 1.0 **Owner:** Solo founder **Status:** Pre-build **Companion document:**

SOFTWARE_ARCHITECTURE_AND_TECHNICAL_DESIGN.md

1. Summary

Sijil is an AI-driven trade-compliance platform for UAE importers, exporters, freight forwarders, and customs brokers. It ingests shipment documents (commercial invoice, packing list, bill of lading), predicts the correct 12-digit UAE HS code for each line item, catches the cross-document inconsistencies that trigger Dubai Customs (Mirsal 2) Yellow/Red-channel review, flags which UAE category-specific permits apply to a shipment before it's booked, and determines whether a shipment qualifies for preferential tariff treatment under a UAE CEPA agreement.

2. Problem Statement

UAE trading companies lose time and money at four recurring points in the shipping process:

1. **Misclassified HS codes** — a genuinely hard, error-prone task done manually today, with direct consequences for duty calculation and which permits apply
2. **Cross-document mismatches** — invoice, packing list, and bill of lading are prepared somewhat independently and often disagree on quantities, values, or names, which is one of the most common causes of Mirsal 2 flagging a shipment for manual review or physical inspection
3. **Missed or late-discovered permit requirements** — ~10 different UAE regulators (MOCCAE, MOHAP, TDRA, ESMA/MoIAT, Ministry of Economy for strategic goods, and others) each cover different product categories, and companies often discover a permit is required only after a shipment is already in transit
4. **Unclaimed CEPA duty savings** — the UAE has signed 20+ trade agreements in recent years, each with its own rules-of-origin requirements; most companies don't have the in-house expertise to determine if a shipment qualifies, so many simply don't claim preferential treatment they're entitled to

No existing product solves all four together for the UAE market specifically. Global AI classification tools aren't localized to Mirsal 2 or UAE's permit landscape. UAE freight-forwarding software treats documents as storage, not compliance intelligence. DP World's CARGOES platform is the most credible existing competitor but is built as broad, government-integrated logistics infrastructure sold B2G2B — not a focused, self-serve compliance tool for a mid-size trading company.

3. Why This Project Exists

To reduce customs delays, fines, and duty overpayment for UAE trading companies, and to give customs brokers a force-multiplying tool — without requiring a government license or lengthy certification process to start selling, which rules out categories like the OT/ICS-security business considered earlier in this research process.

4. Target Users

- **Primary:** SME and mid-size trading companies in the UAE (roughly 5–500 employees) who import or export goods regularly
- **Secondary:** Customs clearing agents and freight forwarders acting on behalf of multiple client companies
- **Tertiary (future):** Larger enterprises wanting API-level integration into their own ERP systems

5. Goals

Business goals

- Reach a working MVP that a solo technical founder can realistically build and operate
- Land the first 10 paying pilot customers (trading companies or brokers) without needing government approval to start selling
- Validate that the product measurably reduces Yellow/Red-channel incidents and missed-permit incidents for pilot customers

Product goals (MVP — full scope, as decided)

- Predict HS codes with a visible confidence score and human-readable reasoning, never a silent guess
- Catch document inconsistencies before the user files with Mirsal 2

- Tell the user which of the ~10 UAE regulators require a permit for their shipment, before it's booked
- Determine CEPA preferential-tariff eligibility for a curated set of the UAE's highest-volume trade agreements
- Produce a shareable compliance report per shipment

6. Non-Goals (Out of Scope for MVP)

- Direct Mirsal 2 filing/submission — Sijil is advisory; the user still files themselves at MVP
- Paying duties/VAT on the user's behalf
- Freight booking, carrier rate comparison, or container tracking (DP World CARGOES / QuickMove territory, not the wedge)
- Full ERP or accounting functionality
- Jurisdictions outside the UAE
- Physical document notarization/legalization workflows

7. Key User Stories

1. *As a trader*, I want to upload my invoice and packing list before booking a shipment, so I know the correct HS code and my confidence in it before I commit to shipping.
2. *As a trader*, I want to be warned if my invoice and packing list disagree on quantity or value, so I can fix it before customs flags my shipment.
3. *As a trader*, I want to know which government permits my shipment needs, before I've already paid for freight, so I'm not caught off guard mid-transit.
4. *As a trader*, I want to know if my shipment qualifies for a reduced tariff under a UAE trade agreement, so I don't overpay duty I didn't have to.
5. *As a customs broker*, I want to manage compliance checks across multiple client companies from one dashboard, so I can serve more clients without proportionally more manual work.
6. *As a company admin*, I want to control which of my team members can view or edit shipment data, so sensitive commercial information stays appropriately restricted.

8. Success Metrics

- **Product quality:** HS classification accuracy against a maintained golden-set of verified product→code pairs (tracked continuously, per the architecture doc's Section 21 testing strategy)
- **User outcome:** Reduction in Yellow/Red-channel incidents reported by pilot customers using Sijil vs. their prior baseline
- **Adoption:** Number of paying companies/brokers onboarded; shipments processed per month
- **Trust signal:** Manual-review rate (percentage of classifications/permits/CEPA determinations the system itself flags as needing human review) — this should be transparent and trending in a healthy direction, not hidden

(Specific numeric targets are intentionally left open here — see Section 12 — since no real usage data exists yet to set them credibly.)

9. Assumptions

- HS classification is offered as an AI-assisted advisory suggestion, not a legally-binding classification service, until the product has traction and appropriate legal/insurance coverage is in place
- Permit-applicability rules are maintained as a curated, versioned internal dataset rather than scraped live from government sites at request time
- CEPA coverage starts with a curated subset of the UAE's highest-volume trade agreements (e.g., India, Indonesia) and expands over time
- UAE/GCC-region hosting is preferred for data residency comfort, though not yet confirmed as a strict legal requirement for this specific data category

10. Risks

- **Scope risk:** building all four capabilities (classification, consistency, permit triage, CEPA) simultaneously as a solo founder is ambitious — mitigated by the phased build order in the architecture document (classification first, as the core differentiator; permit-rules data collection in parallel since it's research work, not blocked by engineering)
- **Data maintenance risk:** permit and CEPA rules require ongoing manual upkeep as regulations change — this is an operational commitment, not a one-time build, and is the single biggest non-technical risk to the product's long-term credibility
- **Competitive risk:** DP World's CARGOES platform has far deeper government

integration and capital; Sijil's defensibility depends on staying focused on the self-serve, AI-driven compliance-intelligence niche CARGOES doesn't serve well, rather than trying to compete on breadth

- **Unit economics risk:** LLM API cost per shipment is a meaningful cost driver that needs real-usage validation early

11. Dependencies

- An LLM provider capable of structured-output, reasoning-heavy tasks (document extraction, GRI-based classification reasoning)
- An OCR service for scanned/image document ingestion
- A billing provider supporting UAE-relevant payment methods
- The founder's ongoing research/maintenance of the permit-rules and CEPA-rules datasets — a non-engineering dependency worth planning for explicitly

12. Open Questions

- What are the exact legal requirements around data residency and retention for UAE trade-compliance data? (Needs legal consultation before finalizing.)
- Is any liability or insurance structure needed before offering classification advice commercially?
- Which specific CEPA agreements should be prioritized first — this should be validated against actual pilot customers' trade lanes rather than decided speculatively.
- What pricing model fits best — per-shipment, flat subscription, or tiered by company size? Needs validation with early pilot conversations.
- What are realistic, evidence-based numeric targets for the success metrics in Section 8, once initial usage data exists?

13. Relationship to Other Project Documents

This PRD defines *what* is being built and *why*. For *how* it's built — system architecture, data model, API contracts, and the phased implementation plan — see

`SOFTWARE_ARCHITECTURE_AND_TECHNICAL_DESIGN.md`, which was written first and from which this PRD has been extracted and expanded (its Section 1 originally stood in for this document).

docs/

— PRD.md	← this document
— SOFTWARE_ARCHITECTURE_AND_TECHNICAL_DESIGN.md	← how it's built
— API_SPEC.md	← not yet written; architectu
— IMPLEMENTATION_PLAN.md	← not yet written; architectu